

Trainings ▾

With Mechanical or Electrical Metering Equipment (Ether)

⚠ WARNING ⚠

Do not use starting fluids on engines in marine applications. Starting fluid is highly flammable and explosive. Keep flames, sparks, and arcing switches away from starting fluid.



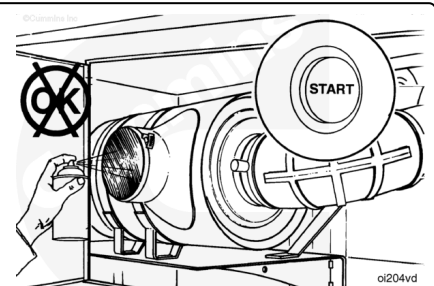
⚠ WARNING ⚠

Do not use volatile cold starting aids in marine applications due to the potential of an explosion.

⚠ WARNING ⚠

Because of the potential for an explosion, do not use volatile cold starting aids in underground mine or tunnel operations. Ask the local U.S. Bureau of Mines inspector for instructions.

Due to the increased safety hazards and potential for engine damage, do **not** use starting fluid without metering equipment.

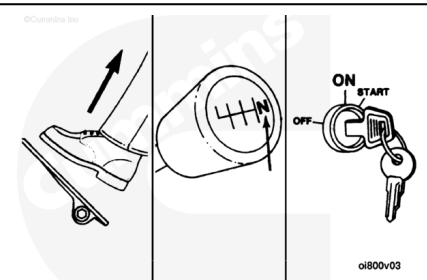


Set the throttle at idle.

Disengage any driven accessories, and, if equipped, shift the transmission in neutral.

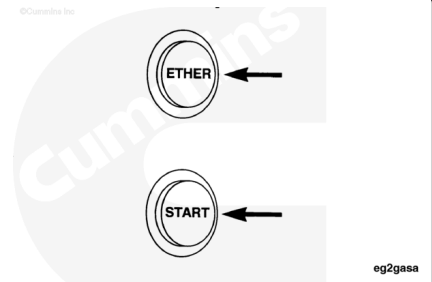
Note : On engines equipped with a PRELUB™ system, hold the key in the START position to allow lubrication oil pressure to build up.

Turn the keyswitch to the ON position to power up the engine control module (ECM).



⚠ CAUTION ⚠

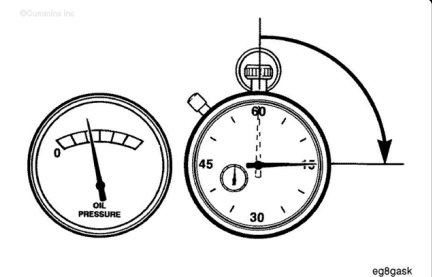
Do not use excessive amounts of starting fluid when starting an engine. Using too much starting fluid can cause engine damage. On engines equipped with a PRELUB™ system, do not use ether until the engine begins to crank.



While cranking the engine, inject a metered amount of starting fluid.

⚠ CAUTION ⚠

The engine must have adequate oil pressure within 15 seconds after starting. If the warning lamp indicating low oil pressure has not gone out or there is no oil pressure indicated on the oil pressure gauge within 15 seconds, shutoff the engine immediately to reduce the possibility of engine damage.



Confirm the correct oil level in the lubricating oil pan. Refer to Procedure 007-043 in Section 3.

([/qs3/pubsys2/xml/en/procedures/99/99-007-043-om.html](http://qs3/pubsys2/xml/en/procedures/99/99-007-043-om.html))

On engines equipped with a PRELUB™ system, the starter will **not** engage until adequate oil pressure is obtained.

Do **not** increase the engine speed above low idle until the coolant temperature gauge starts to move and minutes have elapsed. This will provide adequate lubrication to the bearings.

Monitor the oil pressure gauge after normal operation is initiated.

